Comparative Assessment: MILU vs. MMLU

Deployment context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Deployment Context

Use case: Deployment scenario: A Hindi-speaking student in India is testing his/her preparation for competitive job-related examinations, with an AI system providing feedback on their responses. The goal is to evaluate the applicability of the benchmark and the quality of the LLM’s feedback.

Domain: Educational assessment

Setting: Mobile application / enterprise software

Note: The deployment hypothesis should be tested using Hindi-language sentences from the benchmark, as I am familiar with the language and can provide evaluation in a subsequent stage.

Target population: A graduate student in North India interacting in the Hindi language. The student has completed their education primarily in Hindi, with limited exposure to English.

Overall Risk Ratings

Benchmark	Risk	Avg. Score
MILU	HIGH	2.2 / 5
MMLU	HIGH	1.2 / 5

Dimension Scores Comparison

Dimension	MILU	Rating	MMLU	Rating	Δ
Input Ontology (IO)	3/5	Moderate gaps	1/5	Serious concern	+2
Input Content (IC)	2/5	Significant gaps	1/5	Serious concern	+1
Input Form (IF)	4/5	Minor gaps	1/5	Serious concern	+3
Output Ontology (OO)	1/5	Serious concern	1/5	Serious concern	0
Output Content (OC)	2/5	Significant gaps	1/5	Serious concern	+1
Output Form (OF)	1/5	Serious concern	2/5	Significant gaps	-1
Average	**2.2**		**1.2**		**+1**

Δ = MILU score − MMLU score. Positive = regional benchmark scores higher.

Overall Summaries

MILU (risk: HIGH)

MILU is the most relevant publicly available benchmark for Indic-language knowledge evaluation and provides genuine value for a portion of this deployment's needs — particularly for measuring MCQ-level accuracy on Indian Polity, History, Geography, and Economics in Hindi. Its India-first design [Q11], real-exam sourcing [Q24, Q25], and 41-subject taxonomy [Q42] are well-suited to a central-exam preparation context in principle. However, two structural issues sharply limit its validity for THIS deployment: (1) Output form/ontology mismatch (Output Ontology=1, Output Form=1) — MILU evaluates only MCQ option selection, while the deployment requires open-ended Hindi explanatory rationales; this is a fundamental scope gap acknowledged by the paper itself

[Q85] and confirmed by data inspection (DATASET-D1, D9). (2) Hindi partition content concerns (Input Content=2) — dataset analysis found 100% is_translated=True in the 26-item Hindi validation sample (DATASET-D13), heavy Engineering/Tech skew misaligned with UPSC/SSC syllabi (DATASET-D5/D6/D7/D16/D36), Current Affairs staleness (DATASET-D4/D25/D33), and pervasive cross-language item duplication. Input form (4) is the strongest dimension; output ontology and output form are the weakest. The benchmark can serve as a partial accuracy probe for a subset of priority subjects but cannot validate the deployment's core output.

MMLU (risk: HIGH)

MMLU is a near-total mismatch for the North India Hindi-medium competitive exam deployment across five of six validity dimensions. Subject coverage is anchored in US/Western academic curriculum with zero Indian Polity, Indian History, Indian Geography, Current Affairs, or Hindi-language content (the high-priority domains identified in elicitation). Individual items are culturally embedded in US institutional, legal, demographic, and moral contexts; where India does appear, it is filtered through colonial or Western-pollster framing incompatible with the NCERT/Bipin Chandra-aligned exam-prep tradition. The benchmark is entirely English with no Devanagari content, conflicting with the deployment's Hindi-dominant input requirement. Output is label-only with no rationale evaluation, while the deployment requires substantive Hindi explanatory feedback. Annotator pool is undocumented but US-anchored; moral_scenarios labels are explicitly framed in US 2020 norms. Output form is the only dimension with a partial structural match (MCQ format), but even that is undermined by calibration failures and absence of generative-quality measurement. Indian-grounded alternatives (MILU, ParamBench, IndicEval, IndicMMLU-Pro) exist and should be primary; MMLU at best provides weak proxy signal on STEM/reasoning subdomains.

Per-Dimension Justification Comparison

Each benchmark's full assessment justification and cited evidence, shown side by side.

Input Ontology (IO)

Whether the benchmark's test case categories cover the query types expected in deployment.

MILU — 3/5 (Moderate gaps) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
MILU's 8 macro-domain / 41-subject taxonomy [Q42] aligns partially with central-exam priority subjects: Social Sciences and Law & Governance map to History/Polity, while Arts & Humanities covers cultural GK. However, the taxonomy does not separately enumerate Mathematics/Reasoning or Current Affairs as named categories — both heavily weighted in UPSC CSAT and SSC CGL Tier 1 [WEB-9, WEB-10, WEB-15]. Dataset analysis confirms reasoning and quantitative items exist (in the English partition: D8, D9), but the Hindi sample shows a heavy skew toward translated Engineering & Tech items (DC motor, half-wave rectifier, transformer, AM bandwidth, Fortran 77) that are peripheral to UPSC GS / SSC GA syllabi. State-level and central-level items are pooled without metadata distinguishing them [Q27], making it impossible to filter for central-exam scope (D24, D27).	MMLU's 57-subject taxonomy is anchored in a US/Western academic curriculum [Q1, Q8, Q16] with subjects calibrated to US bar-exam, USMLE, and US-foreign-policy concerns [Q17, Q49]. None of the high-priority Indian competitive exam domains identified in the elicitation (Indian Polity & Constitution, Indian History, Current Affairs, Indian Geography, Hindi language proficiency) is represented. Dataset analysis confirms across ~200 sampled items zero coverage of Indian Polity, Indian History, or Indian Geography; the only South Asian content is two Jainism items in world_religions and one colonial-framed 1857 item. Partial overlap exists for STEM/mathematics and reasoning, which are universally applicable, but this is a small fraction of the deployment's high-priority surface.

MILU evidence	MMLU evidence
[Q42] 'Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.' (p.4)	[Q1] 'The test covers 57 tasks including elementary mathematics, US history, computer science, law, and more.' (p.1)
[Q27] 'Our benchmark includes questions from over 40 different types of exams conducted both at the national and state levels over recent years.' (p.3)	[Q8] 'The benchmark covers 57 subjects across STEM, the humanities, the social sciences, and more.' (p.1)

[WEB-9] UPSC Prelims subject weightage — Current Affairs, Economy, History, Geography, Polity, Environment each 10–25 questions/year	[Q17] 'practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination' (p.3)
[WEB-10] SSC CGL Tier 1 covers Reasoning, General Awareness, Quantitative Aptitude, English Comprehension	
[WEB-15] UPSC CSAT Paper 2 covers Quantitative Aptitude and Logical Reasoning (qualifying)	
DATASET-D8: Logical Reasoning analogy item present in English partition — confirms reasoning subject exists in MILU taxonomy	
DATASET-D9: Quantitative aptitude (compound interest) in English partition — confirms math subject coverage	
DATASET-D24: Maharashtra Legislative Committee question — state-level governance content not in central-exam scope	
DATASET-D27: MP industrial geography question — state-level content commingled with central-exam items	

Input Content (IC)

Whether individual datapoint content is culturally and contextually appropriate for the target region.

MILU — 2/5 (Significant gaps) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
MILU's stated design priorities — India-first sourcing, regional cultural content, sourcing from real exam papers [Q11, Q12, Q24, Q25] — are well aligned with the deployment's content needs. However, dataset analysis reveals two critical issues for the Hindi partition: (1) all 26 Hindi validation examples carry is_translated=True (DATASET-D13), suggesting the Hindi partition may contain a much higher translated proportion than the documented overall ~25% [Q47]; (2) cross-language item duplication is pervasive (DATASET-D14/D15/D29), reducing effective Hindi-specific content. North India sub-regional content is sparse in the observed Hindi sample (only DATASET-D3 on UP minerals). Current Affairs items are time-bounded to 2014–2022 events (DATASET-D4, D25, D33), creating staleness risk for 2025–26 exam cycles. While translated items use fluent Devanagari with low code-mixing (DATASET-D12, D21), the linguistic register may not match authentic Hindi-medium exam papers.	Individual MMLU items are culturally anchored in US/Western contexts: US Congress and constitutional history [DATASET-D1, D2, D3], US bar-exam tort scenarios [DATASET-D10, D11], Western philosophical canon [DATASET-D16, D17], US Social Security and birth-control demographics [DATASET-D20, D21], US measurement units in mathematics [DATASET-D25, D28], and CPA/MCAT institutional referents [DATASET-D32, D48]. The benchmark assumes models acquired requisite knowledge from 'vast quantities of diverse Internet text' [Q60], which is overwhelmingly English-Western. The deployment requires both pan-India and North India–specific factual content (Chhath Puja, Panchayati Raj, tehsils, NCERT-aligned narratives) per elicitation Q2, none of which appears in the sampled items. Where India is referenced, it is via Pew Research statistics [DATASET-D23, D24] or colonial-British framing of the 1857 uprising [DATASET-D15] — both incompatible with NCERT/Bipin Chandra-aligned exam preparation.

MILU evidence	MMLU evidence
[Q11] 'We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams.' (p.2)	[Q60] 'we assume that models have acquired the requisite knowledge from reading vast quantities of diverse text from the Internet' (p.7)
[Q24] 'collecting the questions from various public exams taken by individuals intending to either pursue higher studies or seek career advancements, such as qualification tests and national and state-level civil services exams' (p.3)	DATASET-D10, D11, D33: US tort/contract law scenarios

[Q25] 'We gathered exam-specific questions by scraping various online exam portals that offer previously released question papers from various exams in multiple different languages.' (p.3)	DATASET-D15, D47: 1857 uprising rendered through colonial British journalist's lens
[Q47] 'Of the total 79K questions, only 25% of questions are translated from English, with the remainder' (p.4)	DATASET-D25, D28: US customary units (miles, feet) in arithmetic
[Q12] 'These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science.' (p.2)	DATASET-D23, D24: India-referencing items source Pew Research statistics rather than UPSC GK conventions
DATASET-D13: All 26 Hindi validation examples carry is_translated=True — sample suggests Hindi partition translation rate substantially exceeds the documented 25% overall average	
DATASET-D14/D15: Identical 1991 financial crisis question appears in English original and Hindi translation — confirms cross-language translation pipeline	
DATASET-D29: 44th Constitutional Amendment question appears identically across Hindi, Tamil, Malayalam — cross-language item duplication reduces Hindi-specific diversity	
DATASET-D3: UP minerals question — only observed North India sub-regional item in Hindi sample (1 of 26)	
DATASET-D4/D25/D33: Current Affairs items reference 2022 events — stale for 2025–26 exam cycles	
DATASET-D12/D21: Hindi items use clean Devanagari with minimal English intrusion — code-mixing within deployment ceiling	

Input Form (IF)

Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

MILU — 4/5 (Minor gaps) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
MILU is text-only with explicit exclusion of reading-comprehension passages, image-based items, and >4-option items [Q33], plus INDICLID + Unicode-based language filtering [Q34], duplicate removal [Q35], and multi-stage manual cleaning [Q31, Q32, Q36]. This matches the deployment's text-only mobile interface and standard Devanagari script with no RTL/capitalization concerns. Observed Hindi items render in standard Devanagari with minimal Roman intrusion (DATASET-D12, D21). One residual concern: dataset analysis observed at least two malformed items with truncated stems across language partitions (DATASET-D34, D35), suggesting filtering imperfections.	MMLU is entirely English-language, text-only, with a standardised English preamble for prompts [Q44] and English-only normalised input format for UnifiedQA [Q91]. Dataset metadata confirms 'languages: [en]' and monolingual content; no Devanagari script or Hindi vocabulary appears in any of the ~200 sampled items. The deployment requires Hindi-dominant interaction in Devanagari script with at most ~10% English code-mixing per elicitation Q4. The MCQ structural format (4 options A/B/C/D) is a partial structural match to UPSC/SSC exam formats, but this is overridden by total language and script mismatch. IndicEval [WEB-16] empirically demonstrates a large Hindi-vs-English performance gap (LLaMA 39.53% on UPSC Hindi Zero-Shot vs. ~84.88% Gemini on UPSC English CoT), confirming that English-medium MMLU scores cannot predict Hindi-medium performance.

MILU evidence	MMLU evidence

[Q33] 'During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency.' (p.4)	[Q44] 'We begin each prompt with "The following are multiple choice questions (with answers) about [subject]"... up to 5 demonstration examples... All prompts end with "Answer: "' (p.6)
[Q34] 'To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language.' (p.4)	[Q91] 'questions and choices are normalized and made lowercase' (p.12)
[Q31] 'we employ multiple layers of manual and automated cleaning filters.' (p.4)	[WEB-16] IndicEval documents marked accuracy drops in Hindi vs. English (LLaMA 39.53% UPSC Hindi Zero-Shot)
[Q35] 'we remove any duplicate questions to retain only the unique ones.' (p.4)	
DATASET-D12: Pure Devanagari Hindi item, no English code-mixing	
DATASET-D21: Hindi item with appropriate transliteration of loanwords (ओक/oak)	
DATASET-D34: English item with truncated stem ('The owner of the textile shop brought a') — filtering imperfection	
DATASET-D35: Malayalam item with truncated 'Directions:' stem — pattern repeats across languages	

Output Ontology (OO)

Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

MILU — 1/5 (Serious concern) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
This is a fundamental mismatch. MILU's output ontology is a single MCQ label — one of up to four options [Q33, Q55] — and the dataset schema confirms only a 'target' field with no rationale or explanation field (DATASET-D1, D9). The deployment requires a correct/incorrect verdict PLUS a substantive Hindi-language explanatory rationale (elicitation Q3). MILU has no rubric, no label set, and no scoring function for rationale quality. The 41-subject taxonomy [Q42] supports per-subject accuracy reporting but cannot validate explanation quality. Compounding this, the benchmark itself reports models perform worst in Arts & Humanities, Social Sciences, and Law & Governance [Q6, Q20, Q62, Q71, Q77] — precisely the deployment's priority subjects (History, Polity, GK).	MMLU's output ontology is a four-class A/B/C/D classification [Q83, Q44] with classification accuracy as the sole performance criterion [Q34, Q23]. Dataset analysis confirms a single ClassLabel field (0–3) with no rationale field. The deployment requires a fundamentally richer output taxonomy: a verdict (sahi/galat) plus a substantive Hindi-language explanation in standard Manak Hindi [elicitation Q3]. MMLU's design provides zero signal on the model's capacity to produce explanatory rationale, let alone in Hindi. The benchmark's stated rationale for the simple format ('a simple-to-evaluate test that measures classification accuracy' [Q23]) is internally coherent but structurally incompatible with the deployment's pedagogical-feedback requirement.

MILU evidence	MMLU evidence
[Q33] 'we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency.' (p.4)	[Q23] 'we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions.' (p.3)
[Q55] 'For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability' (p.5)	[Q34] 'we compute the classification accuracy across all examples and tasks.' (p.5)
[Q42] 'we determine 41 distinct subject names, which fall into eight main domains' (p.4)	[Q83] 'Models are fine-tuned to predict one of four classes' (p.12)

[Q6] 'Domain-wise analysis indicates that models perform poorly in culturally relevant areas like Arts & Humanities and Law & Governance compared to general fields like STEM.' (p.1)	
[Q71] 'open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields.' (p.7)	
DATASET-D1: target='option1' is the only output label; no rationale field exists in schema	
DATASET-D9: Compound interest answer is just 'option2' with no worked solution — confirms no explanation field	

Output Content (OC)

Whether ground-truth labels align with the judgments of regional stakeholders.

MILU — 2/5 (Significant gaps) [medium confidence]	MMLU — 1/5 (Serious concern) [high confidence]
For the MCQ-label correctness narrowly, MILU has reasonable provenance: answers are inherited from online exam portals where 'subject experts ensure the accuracy of the answers' [Q26], with AI4Bharat volunteer manual audits [Q87] and researcher review of subject-tag clusters [Q41]. However, the paper reports no annotator demographics, no inter-annotator agreement statistics, and no formal QA protocol beyond manual sampling. Critically, the deployment's core output (Hindi rationale) has no ground-truth labels in MILU at all. Dataset analysis surfaces additional concerns: cross-language item duplication (DATASET-D14/D15/D29) and malformed items with truncated stems (DATASET-D34, D35) suggest QA imperfections at the item level. ~55% of subject labels were assigned via machine translation + GPT-4O-MINI [Q37, Q38], an automated process that may not preserve Hindi-specific terminology fidelity.	The paper is silent on annotator demographics, qualifications, IAA protocols, and label-validation procedures — Output Content concerns are explicitly NOT DOCUMENTED in the benchmark write-up. Questions were assembled by US graduate/undergraduate students [Q17] and validated against US-standard answer keys. No Indian SMEs, UPSC/SSC specialists, or Hindi-medium educators participated. For India-relevant items that do appear (DATASET-D15 1857 uprising), the answer keys privilege colonial historiography over NCERT/Bipin Chandra nationalist framing, directly conflicting with what Indian exam stakeholders would judge correct. moral_scenarios labels are explicitly anchored to 'ordinary moral standards in the US as of 2020' [DATASET-D18, D19], producing systematic OC misalignment for the target population. Even the strongest Indian-grounded benchmarks (ParamBench, MILU) only partially document Indian-SME involvement [WEB-15, WEB-13]; MMLU documents none.

MILU evidence	MMLU evidence
[Q26] 'These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers.' (p.3)	[Q17] 'manually collected by graduate and undergraduate students from freely available sources online' (p.3)
[Q87] 'We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits.' (p.9)	[WEB-15] ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks
[Q41] 'We manually review these clusters and assign appropriate subject labels.' (p.4)	[WEB-13] MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks
[Q37] 'we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information.' (p.4)	DATASET-D18, D19: moral_scenarios explicitly invokes 'US 2020 moral standards' as scoring ground truth

[Q38] 'we first translate the untagged questions into English using INDICTRANS2 (Gala et al., 2023) and then prompt GPT-4O-MINI model to assign an appropriate topic name to the question.' (p.4)	DATASET-D15, D47: 1857 item label privileges colonial framing over NCERT nationalist narrative — direct OC misalignment for Indian deployment
DATASET-D14/D15: Identical 1991 crisis question across English and translated Hindi — labels are correct but duplicated	
DATASET-D29: 44th Amendment question identical across Hindi/Tamil/Malayalam — confirms label propagation via translation	
DATASET-D34: Truncated English stem with assigned label='option2' — label validity questionable for malformed item	
DATASET-D35: Truncated Malayalam 'Directions:' stem with label assignment — same QA gap	

Output Form (OF)

Whether the expected output modality matches regional deployment needs and accessibility.

MILU — 1/5 (Serious concern) [high confidence]	MMLU — 2/5 (Significant gaps) [high confidence]
The output form mismatch is fundamental and acknowledged by the paper itself. MILU evaluates via log-likelihood option selection [Q52, Q53, Q54, Q55] or zero-shot JSON-formatted option choice for API models [Q56, Q57]. The authors note explicitly that this 'may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting' [Q85]. The deployment requires open-ended Hindi-language explanatory generation — a fundamentally different output form. Dataset analysis confirms the schema contains only a 'target' MCQ label field (DATASET-D1, D9) with no infrastructure for evaluating fluency, correctness, or pedagogical quality of free-text Hindi output. External evidence confirms no comparable Hindi-language NLG benchmark for exam rationale exists [WEB-13, WEB-14].	Output form is the dimension with the strongest partial match — both MMLU and the deployment use text-based MCQ as the interaction vehicle, and the binary correct/incorrect verdict can in principle be derived from MMLU's A/B/C/D output [Q83]. However, MMLU evaluates only single-letter classification accuracy [Q34, Q41]; it does not evaluate any natural-language rationale, Hindi fluency, or pedagogical appropriateness. Calibration failures further reduce the value of MMLU's confidence outputs as deployment signal: GPT-3 confidence deviates from accuracy by up to 24% zero-shot [Q57] and 14% few-shot [Q96], with Elementary Mathematics RMS error of 19.4% [Q58]. Given the deployment's negative-marking exam context where confidence calibration matters, this is a real concern. The MODERATE priority weight from elicitation reflects this partial match — score is 2 rather than 1 because the structural MCQ alignment provides some external validity.

MILU evidence	MMLU evidence
[Q52] 'For non-API-based models, we use the LM-EVALUATION-HARNESS' (p.5)	[Q34] 'we compute the classification accuracy across all examples and tasks.' (p.5)
[Q53] 'We use the log-likelihood method, where the probability of a given output string is computed by conditioning it on some provided input' (p.5)	[Q41] 'All values are percentages.' (p.5)
[Q55] 'For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability' (p.5)	[Q57] 'its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects.' (p.7)
[Q56] 'The API-based models are evaluated using the generative approach due to the lack of support for prompt log probabilities.' (p.5)	[Q96] 'gap between accuracy and confidence reaching up to 14%' (p.13)

[Q57] 'We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing.' (p.5)	
[Q85] 'our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.' (p.9)	
[WEB-13] IndicGenBench (ACL 2024) covers Hindi generation but not exam rationale quality	
[WEB-14] arXiv:2508.19831 confirms Hindi instruction-following / conversational benchmarks 'largely unavailable publicly'	
DATASET-D1: Schema contains only 'target' field with MCQ option label — no rationale field	
DATASET-D9: Quantitative answer is just option label, no worked solution — confirms output form is option-selection only	